PV250HE / PV375HE / PV500HE

Operating Manual

EFACEC Engenharia e Sistemas, S.A.
Rua Eng. Frederico Ulrich
APARTADO 3078
4471-907 Moreira Maia
Portugal
Tel. +351 22 943 2000
Fax +351 22 943 2209

service.renewables@efacec.pt

www.efacec.pt

Safety instructions

This manual contains information intended to ensure personal safety as well as to protect the product and connected equipment against damage. The notices referring to personal safety are highlighted in the manual by a safety alert symbol. Notices referring only to equipment damage have no safety alert symbol. Warnings are shown in descending order according to the degree of danger as follows.

⚠	**WARNING** Warnings identify conditions or practices that could result in personal injury or loss of life.

⚠	**CAUTION** Caution identifies conditions or practices that could result in damage to the unit or other equipment.

i	**NOTICE** Notice indicates a situation that can result in property damage if not avoided.

In the event of a number of levels of danger prevailing simultaneously, the warning corresponding to the highest level of danger is always used. A warning that uses a safety alert symbol indicating possible personal injury may also include a warning relating to material damage.

Qualified personnel

Only qualified technical personnel may perform any and all work on the EFACEC Inverters.
For the purpose of the safety information in the Installation instructions, a "qualified person" is someone who is authorized by EFACEC to energize, ground, and tag equipment, systems, and circuits in accordance with established safety procedures.
The associated equipment/system may only be set up and operated in conjunction with this documentation. Commissioning and operation of a device/system may only be performed by qualified personnel.

Disclaimer of liability

The content of this document was checked and corresponds to the described hardware and software. However, since deviations cannot be precluded entirely, we cannot guarantee full consistency. The information given in this publication is reviewed at regular intervals and any corrections that might be necessary are made in the subsequent editions. Also, EFACEC follows a continuous development policy and reserves the right to make changes and improvements in the product described in this document without previous notice.

Index

1 General Information

1.1 Product Description

The PV250HE/PV375HE/PV500HE central inverters convert the photovoltaic solar energy from solar modules into the low or medium voltage power grid.

The power conversion system integrated on these Inverters consists of a robust pulse-width modulated PWM inverter with field proven components and an advanced control system, along with DC and AC switchgears and filtering components.

This range of Inverters integrates an advanced maximum power point tracker (MPPT) that assures optimum power production and maximum yield.

1.2 Use of this Manual

This manual is oriented for installers and operators of photovoltaic systems equipped with PV250HE/PV375HE/PV500HE Inverters. It contains important information and operating instructions for the safe and effective operation, maintenance and possible correction of minor malfunctions of the Inverter. The Operator should be completely familiar with the contents of this manual, and must carefully follow the instructions contained herein.

This document must be stored together with the system documentation and must be accessible at all times

Scope of Validaty

This Manual is valid for the following inverters:

- PV250HE
- PV375HE
- PV500HE

Target group

This documentation contains information of interest of the following users:

- Installation personnel
- Commissioning personnel
- Service personnel

Additional Information

The documents listed below are complementary to this manual:

- PV250HE/PV375HE/PV500HE Installation Manual – Mod. ESER 98-10/07
- PV250HE/PV375HE/PV500HE Communication Protocol – Mod. ESER 100-10/09

- PV250HE/PV375HE/PV500HE Maintenance Protocol – Mod. ESER 107-10/09

2 Description

The PV250HE/PV375HE/PV500HE Inverters are modular and composed by 125 kW units. The next picture presents the master and slave that can be combined to achieve high power ratings.

Figure 1 – Master(left) and Slave(right) units

The master and slave units are very similar and the power stage is the same. However, the master unit comprises some extra hardware like a HMI touch screen interface and an UPS for the control systems power supply.

There are important advantages on this modular concept:

- **Modularity**: is a key factor in this design. If one unit fails, the remaining units will continue to work without any problem. This will increase the availability of the photovoltaic power plant.
- **Efficiency:** the available input power can be conducted to the right number of Inverters, maximizing the total efficiency. Periodically, an algorithm running in the master unit will analyze the total input power on a defined time window and will make the necessary computations to identify the best efficiency point by controlling the number of inverters that are connected to the grid.

The following circuit diagram represents the Modular concept and interconnection between each module. This manual will be focused for the PV500HE but the other models, PV250HE and PV375HE, are very similar.

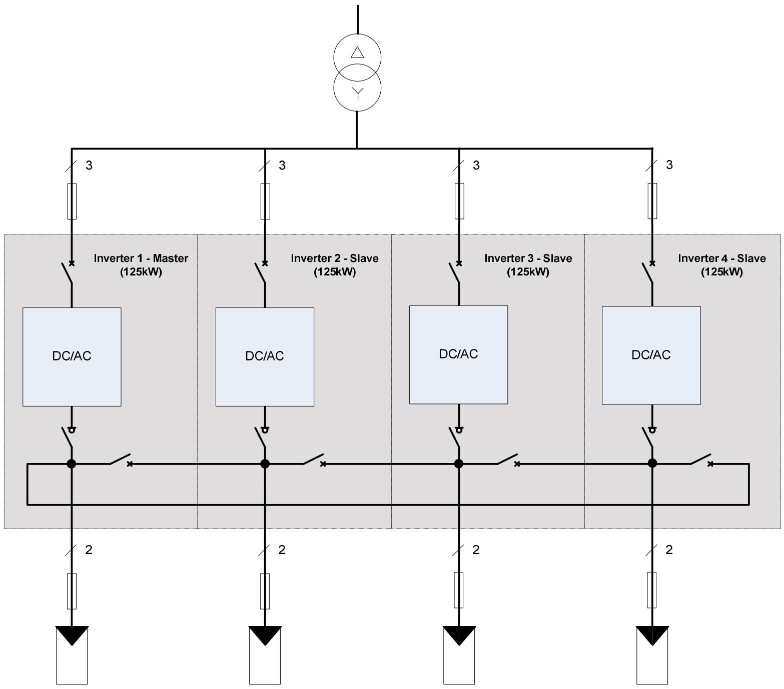

Figure 2 – PV500HE modular concept

As can be seen, for the PV500HE, there are four Inverter Modules of 125kW connected in parallel on the grid side. Each module has is own photovoltaic generator and there is an interconnection bus that allows the distribution of all available power to one or more units.

3 Inverter module

Following the modular architecture, it is important to understand the structure of one Inverter module. In the next sections it will be explained the internal layout of one Inverter module.

3.1 Power circuit

The electrical diagram is represented in the next picture with the most important components identified.

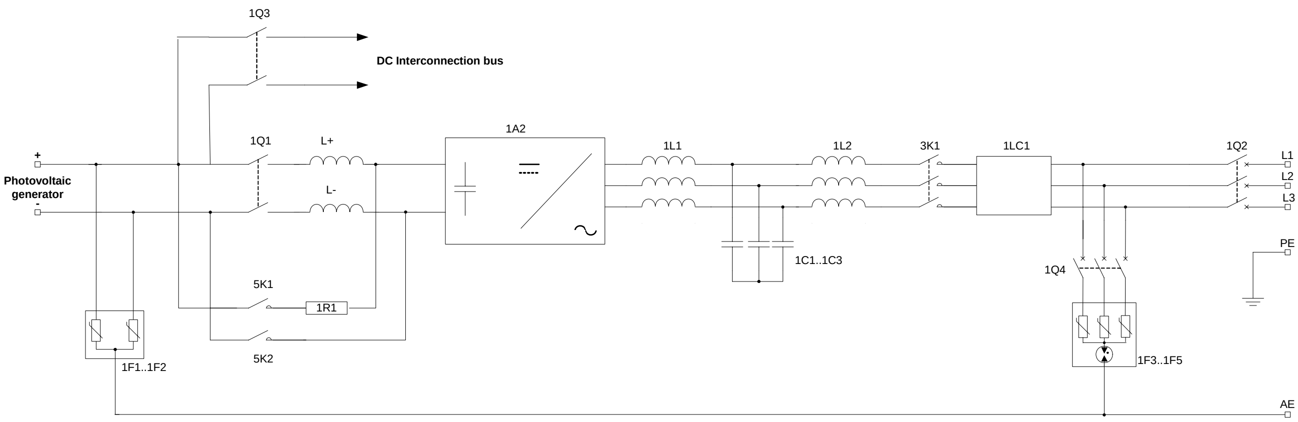

Figure 3 – Inverter Module circuit diagram

Component	Description
1Q3	Interconnection DC breaker
1Q1	Input Switch
F10..F11	Precharge and measuring protection fuses
5K1, 5K2	Precharge Contactors
1R1	Precharge Resistor
L+, L-	DC Input Filter
1F1..1F2	DC Discharger
1A2	Power Stack
1L1	Stack Filter
1L2	Line filter
1C1..1C3	Filter capacitors
3K1	Output Contactor
1LC1	EMI Filter
1Q2	Output breaker
1Q4	AC Surge arresters breaker
1F3..1F5	AC surge arrester

Table 1 – Main Components

The components that are visible from the front side are presented in the next picture. The remaining components can be accessed for inspection by opening the Control Panel door indicated in the picture.

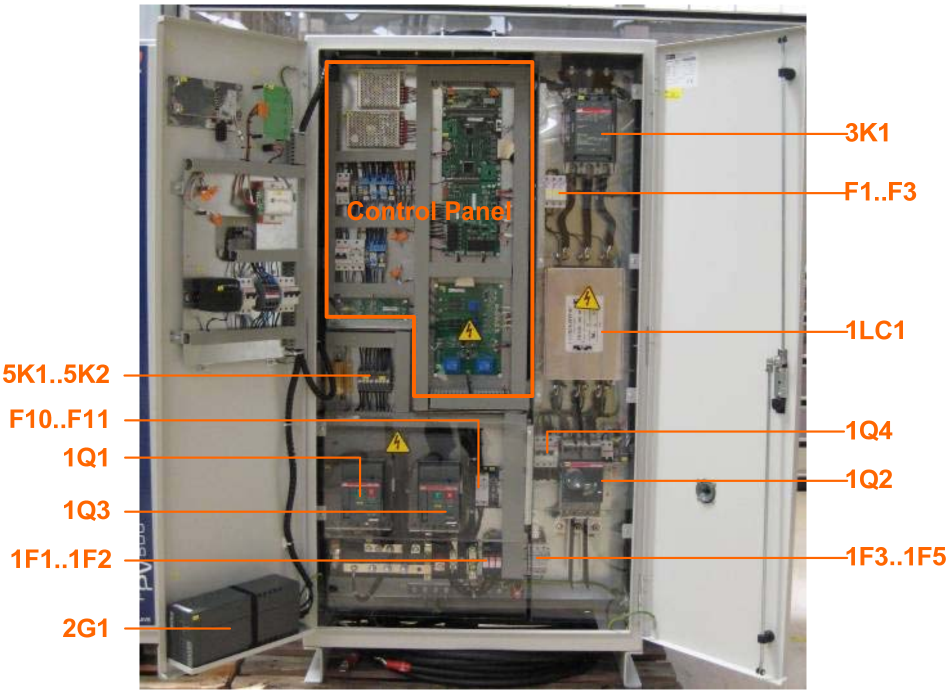

Figure 4 – Inside view (Master unit)

Please note that the 1Q3 switch will be responsible for the connection to the DC bus between the different Modules, allowing the Master/Slave concept.

3.2 Control Panel

The following table shows the list of auxiliary equipment and presents a functional description of its function.

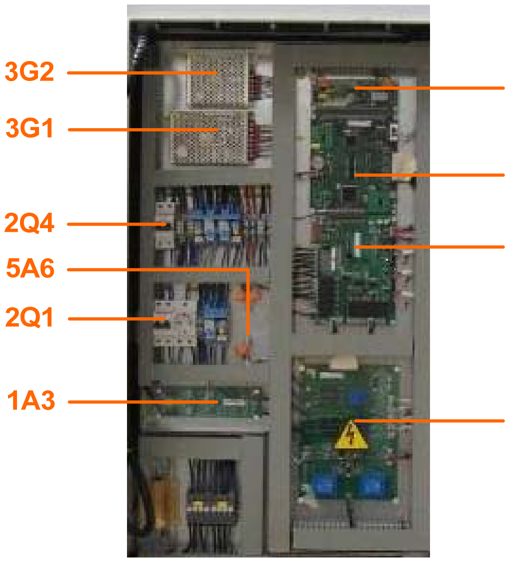

Figure 5 – Control Panel

Auxiliary	Discription
2Q1	Auxiliaries breaker
2Q4	UPS supply breaker
2G1	UPS
3G1, 3G2	Auxiliary power sources
5A6	Communication adapter board
1A3	Earth fault detection board
5A1	Control board
5A2	Digital interface board
5A3	Analog interface board
5A4	Syncronization board

Table 2 – Auxiliaries

3.3 Master / Slave Control unit

As stated, all Inverter Modules are very similar. In fact the major difference between the Master and Slaves is the Master/Slave Control unit installed on the door of the Master module, as can be visualized on the next picture.

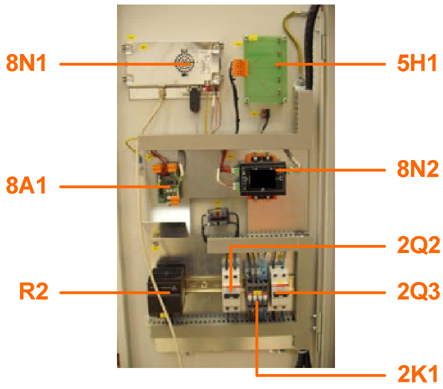

Figure 6 – Master / Slave Control unit

Component	Description
8N1	Master Control unit and HMI Interface
8A1	Communication adapter board
R2	Anti-condensation resistor
5H1	Led panel
8N2	Communication Isolator
2Q2	UPS input breaker
2Q3	Output breaker
2K1	UPS Bypass contactor

Table 3 – Master / Slave Control components

The Master/Slave algorithms are implemented on a powerful embedded computer. This computer has a touch screen interface for the Human Machine Interface (HMI) and also supports the remote monitoring interface.

All individual information from each Inverter Module along with a global view of the system can be accessed, as can be seen in the next sections.

The remaining components are basically used for communication interfacing and power supply for the control systems of all Inverter Modules. There is also installed an UPS to assure the uninterruptible power supply to these systems.

In the case that of any malfunction of the Master/Slave Control unit, the Inverter Modules will work in an autonomous mode, each one with is own photovoltaic generator.

4 Operation

The following diagram represents the operating states of the PV250HE/PV375HE/PV500HE Inverter.

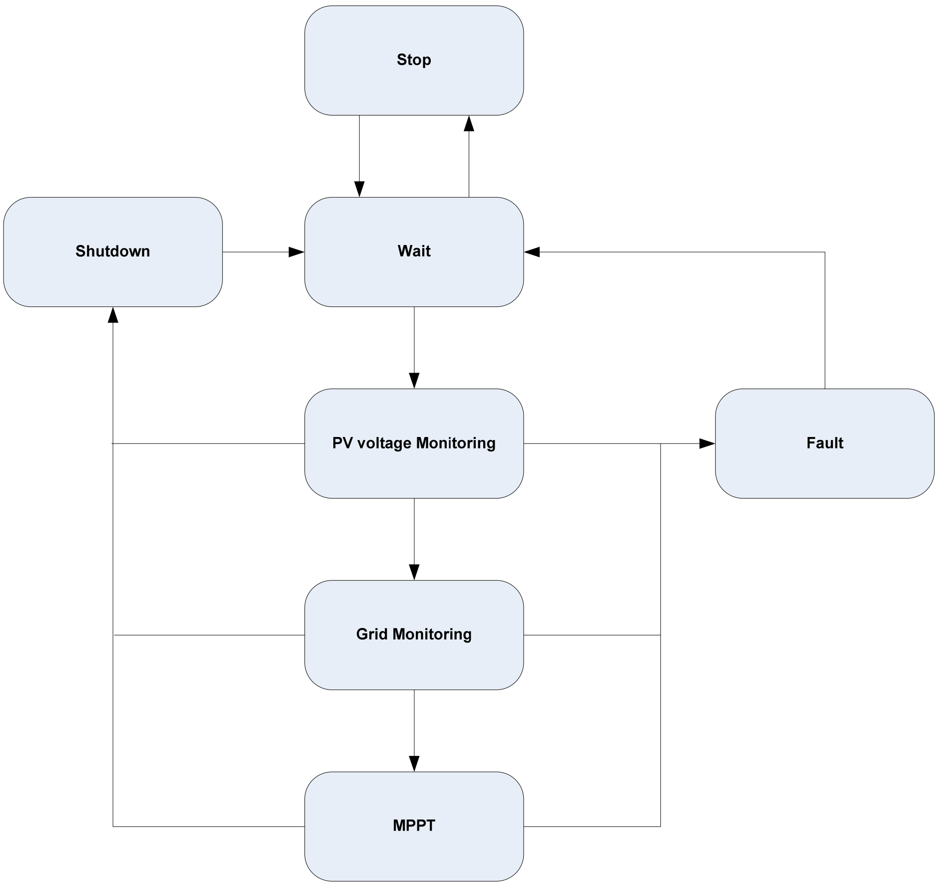

Figure 7 – Operating states

State	Description
Stop	The On/Off button (located on the equipment door) is switched off. The Inverter remains on this state until the button is switched to the "On" position.
Wait	If the switch is set to "On" and Master/Slave Control sends and a Start command (Master mode), the Inverter will wait until the photovoltaic generator voltage value is more than the "UpvStart" limit. On that condition, the input switch is closed. It should be noted that if the Master/Slave Control configures this module as a Slave, the equipment will stay on this state until a start order is sent. In Slave mode, the interconnection switch breaker position is controlled by the Master/Slave Control unit.
Grid Monitoring	After the connection of the input voltage, and during a predefined time, the Inverter will monitor the output voltage and frequency. If these values are inside the expected values, the output contactor will

	close, and the Inverter is connected to the grid.
MPPT	After successful connection of the input switch and the connection to the grid, the Inverter searches for the photovoltaic generator's Maximum Power Point (MPP), and begins feeding into the grid.
Shutdown	The Inverter enters in this state in the following conditions: • If the available input power is less than a preprogrammed value. • If the Master/Slave Control sends a shutdown order. • If the switch On/Off is changed to the "Off" position The shutdown sequence disconnects the Inverter from the grid and from the photovoltaic generator.
Fault	If a fault occurs during operation, the Inverter shuts down. Depending on the type of error, the Inverter can try to ream itself and switch to the Wait operating state. Please note there are critical errors that cannot be rearmed and need local action.

Table 4 – Operating states description

Please note that during the Inverter operation, the Master/Control unit can change the operating mode of each Inverter module and the status of the internal interconnection switch allowing different configurations, as can be seen in Figure 2.

5 Interface Panel

All Inverter Modules have included the User Interface represented on the next picture. The Inverter Slave module interface only has the Led panel and the On/Off switch.

Figure 8 – Master User Interface Panels

Component	Description
1	On/Off switch
2	LED Panel
3	Human to Machine Interface (HMI)
4	Emergency poweroff

Table 5 – Slave and Master User Interface Panels

5.1 On/Off Switch button

This switch button sets the state of equipment as ON or OFF:

State	Description
Off	Inhibition of the inverter - input breaker and output contactor are open
On	Command to connect the inverter. If the input voltage exceeds the programmed value, the input breaker is closed. After a few seconds the output contactor is closed and the inverter starts working. Please note that in the modular configuration, the start order also depends on the Master/Slave Control unit.

Table 6 – On/Off Switch button

In case of failure of the inverter, it can be reset through this switch by simply disable and enable the switch (On position).

5.2 LED Panel

In the door of the equipment is installed a panel with LEDs indicating the following states:

Indicator	Description
On	The inverter is connected and is ready to start to operate.
PV Array	The motorized input switch is closed and therefore the voltage of the photovoltaic modules is present in the power stack.
Grid	The output contactor is closed and the inverter is synchronized with the grid
Failure	Indicates a failure state in the inverter. If the LED is switched On the Inverter will try to do an automatic rearm, otherwise it will be necessary to do a manual rearm to the device, described in the previous section.
Earth Fault	Indicates an insulation fault in the installation

Table 7 –LED Indicators

5.3 Emergency poweroff

The equipment has in the door an emergency poweroff push button. The action triggers the emergency shutdown of all Inverter Modules.

Note that the emergency stop should only be actuated in the case of a critical failure. The equipment will remain in that state until the emergency button is reset.

6 HMI

The PV250HE/PV375HE/PV500HE Inverters are equipped with a graphic HMI on the Master unit. This Interface can be used for the following operations:

- Consult the global information of all Inverter Modules
- Consult the status of each Inverter Module.
- Read the operating states, along with voltages, currents, frequencies, temperatures, output powers, status of errors/warning messages, and events
- Change the hour/date and communication settings

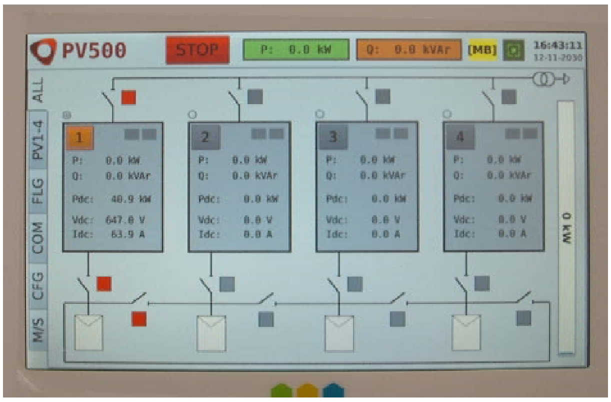

Figure 9 – HMI

As can be seen on the left side of the previous picture, there is a tab selector. It is possible to select different options as explained in the next section.

6.1 HMI menu structure

The HMI menu is structure is represented on the next picture. By default, the Global menu is showed first. In this menu the most important information can be analyzed in this menu.

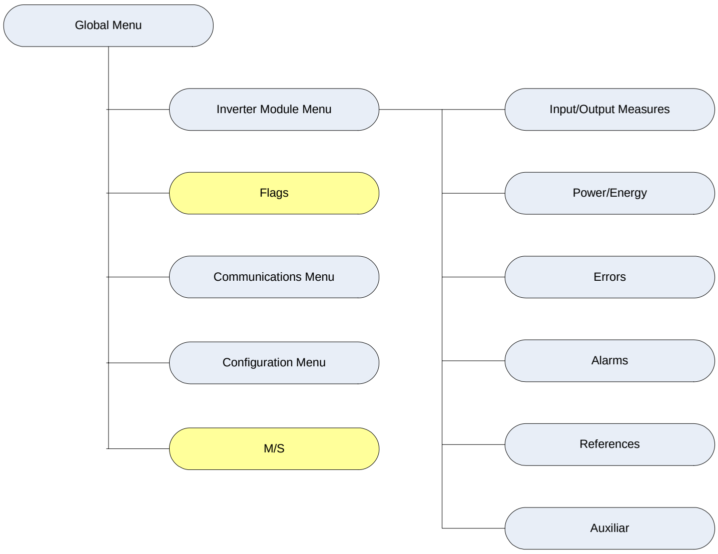

Figure 10 – Menu structure

There other menus, only accessible to EFACEC operators, for maintenance purposes (marked as yellow on the last picture).

6.2 Global menu

This menu presents the Inverter global information, namely:

- Global active and reactive power
- Inverter Module status
- Communication status
- Interconnection switch status

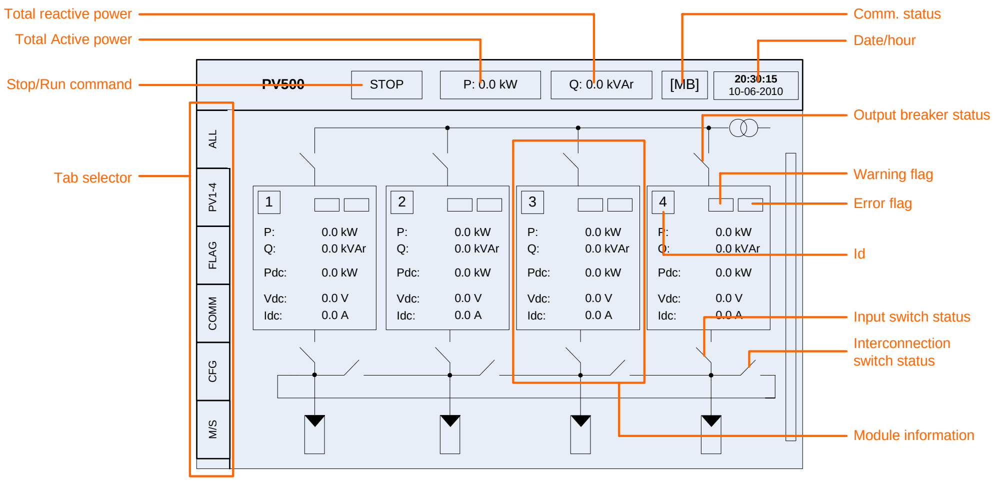

Figure 11 – Global information

From this menu it is possible to jump to the each Inverter module information by pressing the id area.

The information of each module can be easy visualized in this menu. If there is a communication fault between the Master Control unit and the Inverter Modules, the color of the correspondent elements will be grey.

6.3 Inverter Module menu

This menu allows the visualization of individual information of each module. There is a vertical bar where it is possible to select the module address. At the bottom, it is implemented a horizontal bar for the function submenu selection.

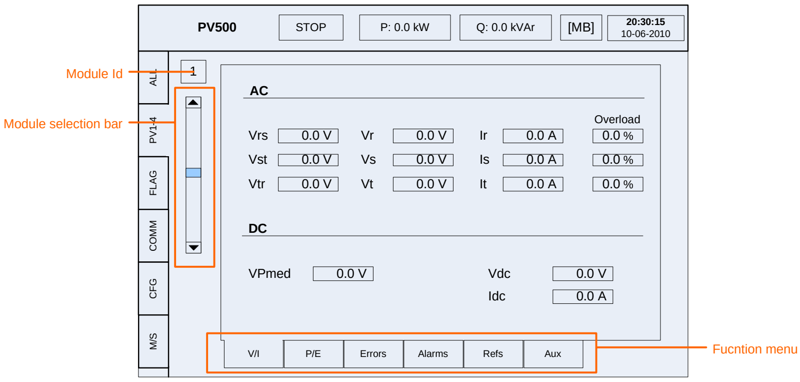

Figure 12 – Inverter Module menu

6.3.1 Input/Output Measures

In the previous picture it is possible to visualize the input and output values, namely:

- Output voltages and currents
- Photovoltaic input voltage
- Input voltage and current
- Overload factors

Please note that the output current sensors are installed on the power stack output (typical configuration of this type of Inverters) and the main function is to control and protect the equipment. This should be taken in consideration when the energy measured by the equipment is used for billing purposes.

6.3.2 Power/Energy

The power and energy generated by the equipment can be consulted on this menu:

- Output power
- Generated energy
- Consumed energy
- Input power
- Power factor

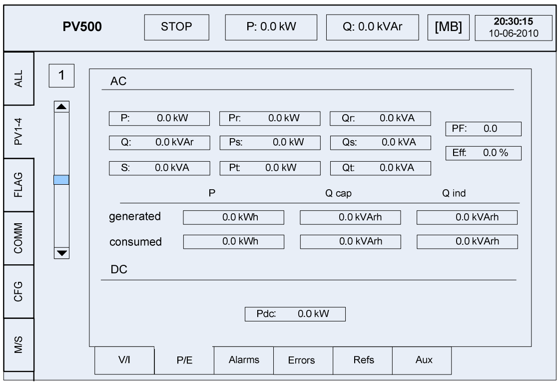

Figure 13 – Inverter Module menu

6.3.3 Errors

The active errors of each Inverter Module can be viewed on this menu. The error information is represented by information flag bits in a color code: red means an active alarm. The correspondence between each bit and the description can be accessed by pressing the “?” button.

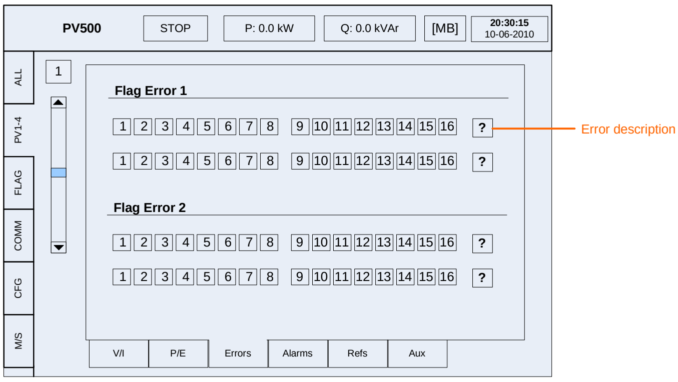

Figure 14 – Error menu

6.3.4 Alarms

This menu summarizes the alarm status. Each bit has an alarm associated and there is a “?” button that can help the translation of each alarm bit.

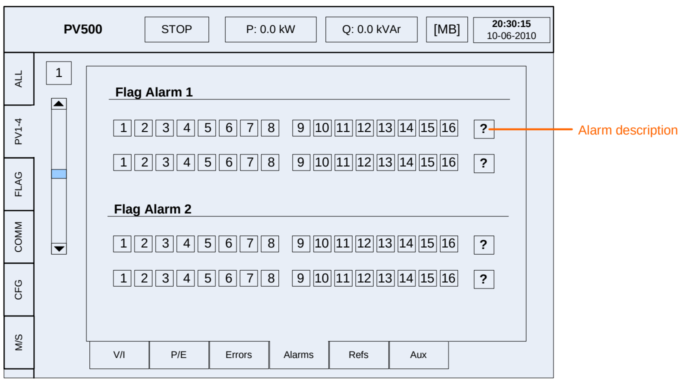

Figure 15 – Alarm menu

6.3.5 References

In this menu is possible to send commands to all Inverter Modules:

- Stop/Run – send a Stop/Run order
- Disconnection – send an order for the DC Switch opening
- Rearm – Clear all errors and sends a rearm order
- Reset Counter – Resets the energy counters on the Inverter Module

It is possible to send the commands to all units by pressing the "All" checkbox prior to send the command.
These operations are possible on the bar signaled on the center of this menu.

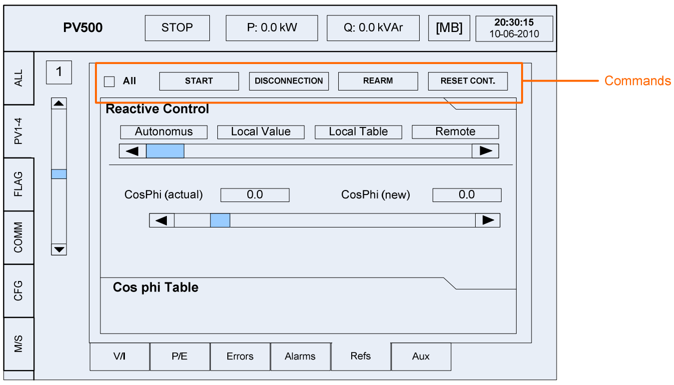

Figure 16 – References menu

The References menu also allows the configuration of the reactive energy compensation mode. There are three possible modes:

- Autonomous (preprogrammed value)
- Fixed *cos φ* value
- Power factor that is configured in the local timetable
- Power factor configured by remote timetable

By pressing the selected option, it is possible to configure the associated information.

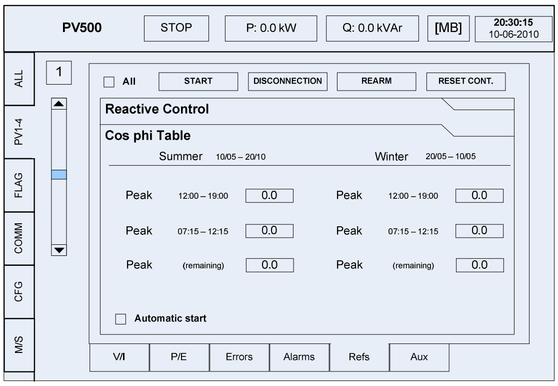

Figure 17 – References

On the option "Local Table", the power factor is programmed locally. It is also possible the select the option "Remote Table" to allow the configuration of the power factor via the communication interface. In this case the table can be consulted on the next menu.

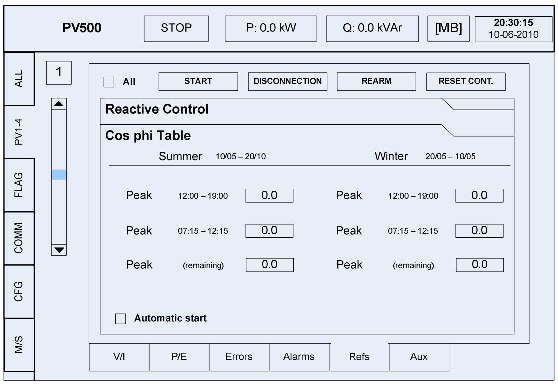

Figure 18 – Remote Table

6.3.6 Auxiliaries

The internal status of each Inverter Module is shown on this menu.

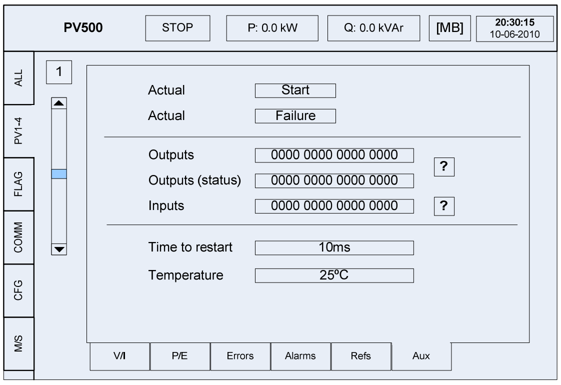

Figure 19 – Auxiliaries menu

Additional information is also available and in particular:

- Time left to restart operation after a fault.
- Module Temperature

6.4 Communication menu

The PV250HE/PV375HE/PV500HE Inverters are equipped with a remote communication interface, using the Modbus protocol directives. On this menu it is possible to configure the SlaveID Address and operation mode (RTU or TCP).

If the selected mode is TCP, the network interface settings must be configured on this menu, by configuring the IP Address and Port (by default is 502).

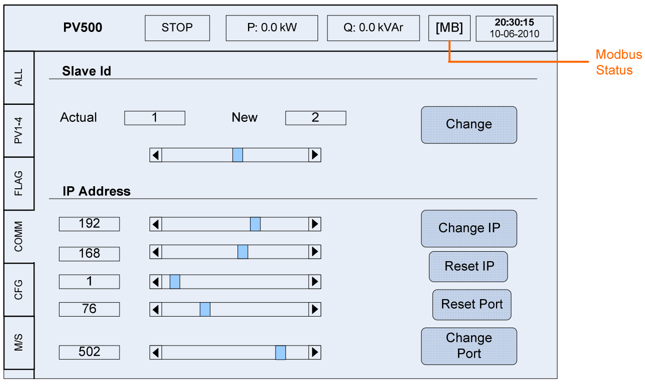

Figure 20 – Communications menu

On the menu header there is a Modbus status indication when a communication is active.

6.5 Configuration menu

Date and hour can be configured on this menu. The version of the HMI Software is also visible on this menu (bottom side).

Figure 21 – Configuration menu

7 Alarm and Error codes

The alarms are grouped into several categories which are associated with a criticality level to each group, and possible action to rearm automatically.

According to the classification of the type of error, is scheduled the following sequence of actions:

Type of error	Action
Normal	Opens the output contactor (3K1)
Severe	Opens the output contactor (3K1), preload contactors (5K1 e 5K2) and input breaker (1Q1)

Table 8 – Type of errors

After detecting an error, the inverter has installed a process of automatic rearm, after 3 minutes. The number of attempts to rearm depends on the type of error but for the majority of cases is 3 times. If the error condition persists, the inverter enters in standby mode and tries the rearm operation after a preset time, according to the next table.

For some errors the associated reset is done on a continuous basis while, for critical errors, the equipment may enter the in the Shutdown status, requiring a locally reset operation.

Error	Description	Reset	Type
ERROR_HIGH_GRID_VOLTAGE	Grid voltage High	1h	Normal
ERROR_LOW_GRID_VOLTAGE	Grid voltage Low	1h	Severe
ERROR_HIGH_FREQ	Grid Frequency High	1h	Normal
ERROR_LOW_FREQ	Grid Frequency Low	1h	Normal
ERROR_GRID_PEAK_VOLTAGE	Grid Voltage Peak	Indefined	Normal
ERROR_PEAK_GRID_CURRENT	Grid Current Peak	Indefined	Severe
ERROR_PEAK_DC_CURRENT	Grid Continuous Grid Peak	1h	Severe
ERROR_OVERLOAD	Overload	Without wait	Normal
ERROR_HIGH_TEMPERATURE	Overtemperature	30 minutes	Severe
ERROR_PRELOAD_DC_LOW_VOLTAGE	Low Preload voltage	Indefined	Severe
ERROR_PRELOAD_LOW_DC_LINK_VOLTAGE	Low DC BUS voltage	Without wait	Severe
ERROR_HIGH_PV_VOLTAGE	High Panels voltage	Indefined	Severe
ERROR_LOW_PV_VOLTAGE	Low Panels voltage	Without wait	Normal
ERROR_HIGH_DC_LINK_VOLTAGE	High DC BUS voltage	Indefined	Severe
ERROR_LOW_DC_LINK_VOLTAGE	Low DC BUS voltage	Without wait	Normal
ERROR_DRIVER	Driver error	Indefined	Severe
ERROR_RESET_DSP	Internal error	Without wait	Normal
ERROR_INPUT_SWITCH	1Q1 auxiliary error	Indefined	Severe
ERROR_MASTER_KCC	Controller error	Indefined	Severe
ERROR_OUTPUT_CONTACTOR	5K3 auxiliary error	Indefined	Normal
ERROR_SURGE_ARRESTERS	Dischargers error	Indefined	Severe

ERROR_GROUND_FAULT	Ground leak error	1h	Severe
ERROR_INT	Controlor error	Without wait	Normal
ERROR_PRELOAD_CONTACTORS	Preload contactor error	Indefined	Severe
ERROR_FAN_SUPPLY	Auxiliary ventilation error	Indefined	Normal
ERROR_LOW_DC_CURRENT	Low IDC current error	1h	Normal
ERROR_LOW_POWER	Low power error	1h	Normal
ERROR_ASYMETRIC_CURRENTS	Asymetric currents error	1h	Normal
ERROR_ASYMETRIC_VOLTAGE	Asymetric voltages error	1h	Normal
ERROR_FAN	Controlor error	Indefined	Normal
ERROR_OUTPUTS	Controlor error	Indefined	Normal

Table 9 – Error list

8 Communications

The master unit integrates a communication controller with a Modbus RTU protocol, available in a RS485 half duplex port. The connection via Modbus TCP protocol is also possible. For more information about the communication or address tables, please consult document PV250HE/PV375HE/PV500HE Communication Interface Mod. ES/ER 100 10/08, or our Technical Support.

9 Maintenance

Once the equipments are installed and are working properly, the maintenance procedure is simple and must be performed periodically to ensure the normal working conditions.

Periodically, it is also necessary to examine the tightness of cables and components of the inverter to ensure proper functioning. Like any electronic equipment, shall be made a regular cleaning in order to avoid the accumulation of dust inside the equipment.

For more information about the Maintenance Protocol, please consult the document PV250HE/PV375HE/PV500HE Maintenance Protocol – Mod. ESER 107-10/09.

10 Environmental Considerations

After the inverter has reached the end of its service life, dispose of it in accordance with the applicable disposal regulations for electronic waste.

The batteries are complying with the indications of Decree-Law No. 62/2001 of 19 February (Council Directive 98/101/EC, the Commission, 22 December).

These batteries can be safely removed at the end of life equipment, allowing the recycling of different components.

The equipment, batteries and accumulators must be delivered in locations/companies authorized. They should not be mixed with other waste.

Type of batteries present on the equipment:

- UPS batteries: Lead (Pb), 12V
- Control Card: Lithium 3V
- User Interface Module: Lithium 3V

ANNEX I - Technical Specifications

		PV250HE	PV375HE	PV500HE
Input	Suggested PV Power	275 kW	412 kW	550 kW
	Input Voltage MPPT	470 V..800 V		
	Maximum Voltage	900 V		
	Maximum Current	595 A	890 A	1200 A
	Maximum Power	275 kW	412 kW	550kW
Output	Nominal Voltage (IT system)	270 V		
	Nominal Current	536 A	807 A	1076 A
	Maximum current	610 A	925 A	1240 A
	Frequency	50 Hz ± 1 Hz		
	Nominal Power	250 kW	375 kW	500 kW
	THD	< 3%		
	Power Factor	>0,98 with nominal voltage and power > 15% from nominal		
Efficiency a)	Maximum	>98,1%	>98,1%	>98,1%
	European efficiency	>97,3%	>97,3%	>97,4%
Protections	Anti-islanding	✓		
	Protection against reverse polarization	✓		
	Transient overvoltages in the input and output	✓		
	Isolation fault in the input	✓		
	Overload	✓		
	Input/Output short-circuit	✓		
	Overtemperature	✓		
	Asymetrical currents	✓		
	Surge Arresters on the a.c. output	✓		
	Surge Arresters on the d.c. input	✓		
Operating Conditions	Temperature	-10º C..+45º C		
	Relative humidity	95% without condensing		
	Ventilation	Forced cooling		
	Fresh air consumption	3000m^3/h	4500m^3/h	6200m^3/h

Auxiliary power supply	Input Voltage (single phase)	230 V		
	Night power consumption	83 W	118 W	169 W
	Maximum power Consumption	670 W	1005 W	1340 W
Interfaces	Synoptic and graphical display with keyboard	PC monitoring software via Modbus		
	Communications			
Mechanical	Width	1800 mm	2700 mm	3600 mm
	Height	1776 mm		
	Depth	710 mm		
	Weight	1210 kg	1820 kg	2430 kg
	Protection degree	IP20		
Approvals	CE Mark	Yes		
	Harmonized Standards	EN61000-6-2, EN61000-6-4, EN50178		
Other features	TCP/IP remote monitoring	○		
	Reactive power compensation module	○		
	Maintenence service	○		
	Warranty extension	○		

[a] Excluding the power supply for auxiliary systems

○ - Optional